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Dual–Continuum Coarse–Grid Network: A Hybrid Approach for Naturally Fractured Reservoir Simulation

Billal Aslam, Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia; Xupeng He and Klemens Katterbauer, Saudi Aramco; Bicheng Yan, Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

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Abstract

Reservoir simulation for naturally fractured reservoirs pose significant challenges due to complex flow behavior driven by fracture–matrix interactions, early fluid breakthrough, and strong heterogeneity. While full-physics fractured reservoir simulation models like the Embedded Discrete Fracture Model (EDFM) offer high fidelity, its relatively high computational cost still limits the applications in iterative workflows like reservoir history matching and optimization. This study introduces the Dual-Continuum Coarse Grid Network (DC-CGNet), a hybrid physics-based data-driven model designed for efficient simulation and history-matching of fractured reservoirs. DC-CGNet employs a coarse dual-layer graph, where matrix and fracture are represented as overlapped graphs. The model parameters in DC-CGNet include relative permeability curves, transmissibility, pore volumes, and the well indices. Due to its coarse discretization nature, the number of model parameters in DC-CGNet is significantly lower than EDFM or traditional Dual-Porosity Models (DPM). The history matching of DC-CGNet can be achieved through coupling with the Ensemble Smoother with Multiple Data Assimilation (ES-MDA). DC-CGNet was examined through a synthetic water-flooding case. The results show that DC-CGNet can accurately history match and forecast dynamic well responses, and achieve up to 15 times speedup. The proposed framework offers a practical, scalable solution for fractured reservoir modeling and history matching, and can be readily integrated into existing reservoir simulators to help guide fast decision-making in reservoir management.

Introduction

Naturally fractured reservoirs (NFRs) play a critical role in global energy production, particularly in carbonate oil reservoirs, tight/shale gas, coalbed methane, and geothermal reservoirs. There widely exist well-connected fracture networks in such reservoirs, and they are typically characterized by Dual-Porosity Models (DPM), where low-permeability matrix stores fluid, and well-connected fracture networks bring rapid flow paths (Barenblatt et al., 1960; Berkowitz, 2002). While fractures can contribute to initial peak production, they often lead to challenges to reservoir management, including early fluid breakthrough, poor sweep efficiency, and uneven displacement fronts, which are difficult to predict and control. As a result, accurate simulation of fractured systems is essential for evaluating reservoir performance, optimizing operational strategies, and supporting data-driven decision-making (Viswanathan et al., 2022). High-fidelity simulation models, such as the Embedded Discrete Fracture Model (EDFM) (Moinfar et al., 2014), can accurately represent the interactions between matrix and fracture. However, their applications in iterative inverse modeling workflows like history matching, uncertainty quantification, and reservoir optimization have still been limited due to the high computational cost, especially for large-scale models. As the uncertainties from reservoir properties, flow physics, and operation settings are often poorly understood or characterized in practice, such inverse modeling tasks can become even more challenging.

Various surrogate models have been proposed to reduce computational burden. Purely data-driven surrogate models based on machine/deep learning approach (Chen et al., 2025; Huang et al., 2023; Yan et al., 2022) can provide extremely efficient predictions once they are well trained, but they often require large training datasets and may still struggle in model generalization. In contrast, physics-based surrogate models can approximate the governing equations or flow interactions with reduced complexity, with examples including such as Proper Orthogonal Decomposition (POD)-based surrogate models (Li et al., 2020; Yang et al., 2016), INSIM (Guo & Reynolds, 2019; Zhao et al., 2016), or streamline simulation (Datta-Gupta & King, 2007). These models are often more robust in physics interpretation, but involve a more complex setup and limited adaptability.

Further, hybrid surrogate models combine physical structure with data-driven calibration (history matching) offer a promising alternative. In particular, graph-based surrogate models have gained increasing attention for their capability to approximate flow dynamics through sparse network including node and edge. Prior works such as FlowNet-DP (Yan et al., 2024) and DiagNet (Aslam et al., 2024) have demonstrated their effectiveness in conventional or sparsely fractured reservoirs. However, hybrid surrogate models for densely fractured reservoirs remains less explored, as the complex and localized flow behavior is difficult to capture through existing graph-based surrogate models.

To address these limitations, this study introduces a hybrid surrogate model designed for densely fractured reservoirs, called the Dual-Continuum Coarse Grid Network (DC-CGNet). Built up on the original CGNet, the DC-CGNet has two sets of CGNet with the same topology to separately represent the matrix and fracture continua, and additional edge are bridged between matrix and fracture nodes to consider the matrix-to-fracture fluid transfer. With its sparse spatial discretization nature, during history matching, DC-CGNet can be parameterized with significantly lower number of parameters than full physics-based EDFM or DPM. Specifically, fracture-to-fracture transmissibilities, fracture-to-matrix transmissibilities, and matrix-to-matrix transmissibilities, matrix/fracture pore volumes, relative permeability curves, and well indices are accounted, with in total less than 1000 parameters. Prior information of fracture distribution is not required to be part of the input variables in DC-CGNet, which makes it a light-weight surrogate. The history matching of DC-CGNet can be performed using the Ensemble Smoother with Multiple Data Assimilation (ES-MDA), and further the calibrated DC-CGNet can be used to forecast the fractured reservoir performance.

Methods

Dual-Continuum Coarse-Grid Network

A typical densely fractured reservoir can be discretized through Embedded Discrete Fracture Model (EDFM) (Moinfar et al., 2014), where the matrix is honored by structured grid cells (e.g., corner point grid) and the network of fractures is explicitly characterized by low-dimensional geometries (lines in 2D or planes in 3D). Since natural fractures can be globally distributed in fractured reservoirs and interacted with matrix through fluid transfer locally, it can be abstracted as the Dual-Porosity Single-Permeability Model or Dual-Porosity Dual-Permeability Model, depending on whether intra-continuum flow can occur in the matrix continuum or not.

The generation of the DC-CGNet follows a similar approach to the original CGNet (Lie & Krogstad, 2023), where a coarse partition of the reservoir domain is constructed to reduce the dimensionality of parameterization, and thus accelerate the computation speed. As shown in Figures 1(a), the DC-CGNet leverages two sets of graphs (nodes and edges) with the same topology to separately represent the matrix and fracture continua. Further, matrix and fracture nodes are bridged by additional edges (dashed lines) to consider the matrix-fracture fluid transfer. Wells are assumed to be only perforated in the fracture graph; thus the well nodes are only connected with the fracture nodes. Therefore, even without fine-scale discretization like standard DPM or EDFM, DC-CGNet can still effectively consider the intra-continuum flow internally in the matrix and fracture, as well as the inter-continuum between the matrix and fracture.

Figure 1(b) illustrate a realistic 3D naturally fracture reservoir model represented by EDFM and DC-CGNet, respectively. The EDFM takes relatively fine mesh to discretize the unstructured fracture network and structured matrix, which makes it relatively computationally expensive. On the other hand, in an equivalent DC-CGNet, the structured matrix discretization is honored yet with relatively coarse graph network (cyan nodes and blue edges), while the unstructured fracture network is abstracted to be equivalent graph network (red nodes and red edges) to matrix graph considering the widely distributed natural fractures.

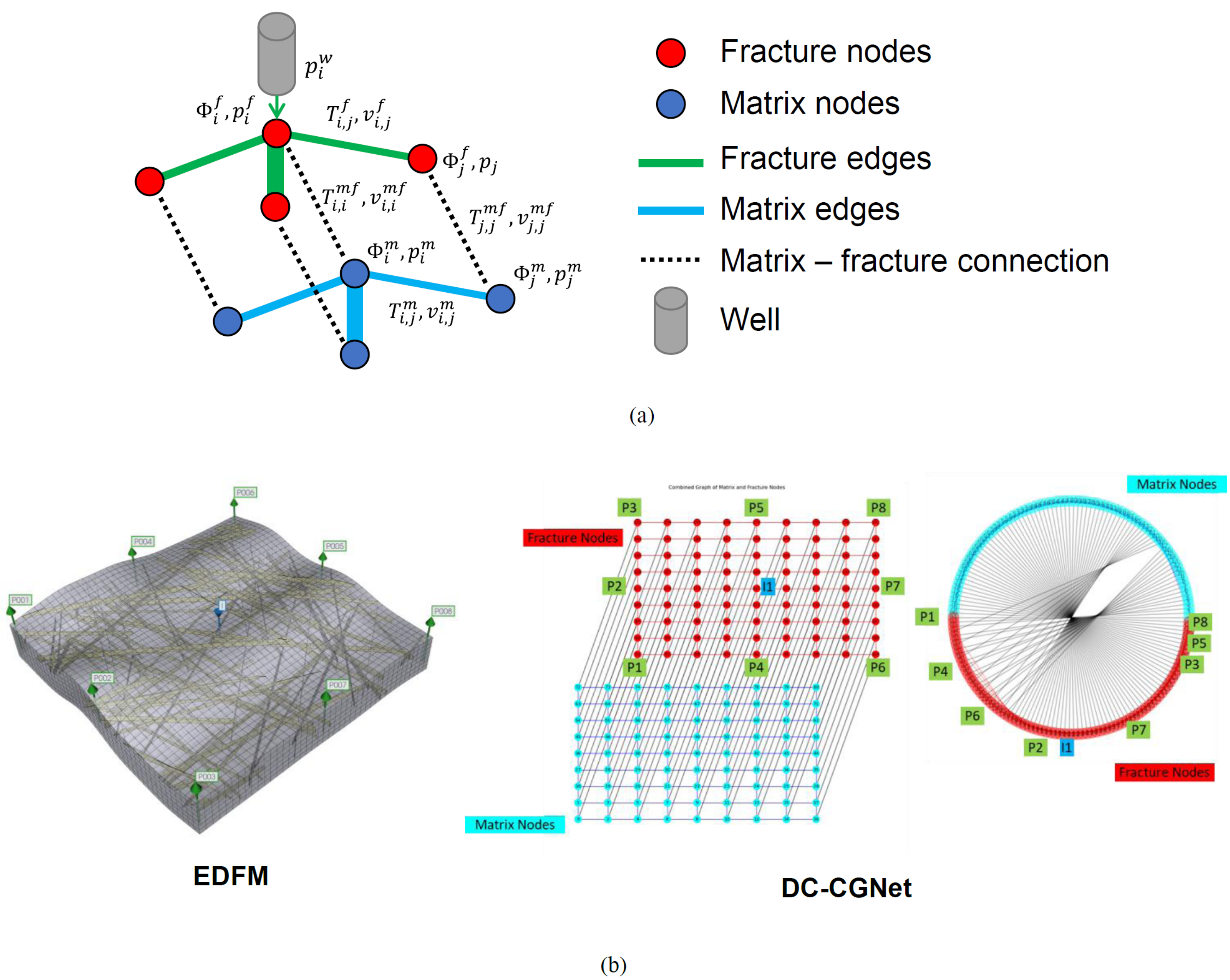


Figure 1—Illustration of DC-CGNet: (a) Concept of DC-CGNet; (b) From EDFM to DC-CGNet.

Remark that the model is compatible with any simulator that supports a connection list for edge properties (i.e., transmissibility), making it easy to implement across different simulator platforms. While the current implementation of DC-CGNet graph uses a dual-layer 2D configuration (Figure 1(b)) for a 3D naturally fractured reservoir, it is able to handle 3D DC-CGNet if necessary. Besides, DC-CGNet can also be customized to unstructured graph, if non-uniform node distribution is required to capture reservoir dynamics.

As DC-CGNet is a spatial discretization technique, it is flexible to further incorporate different physics. To demonstrate the efficacy of DC-CGNet, this work takes an example of water-flooding process in a naturally fractured oil reservoir. The primary considered physics is two-phase flow in a dual-porosity reservoir system, which can be described by the mass conservation law. The mass conservation equations of each fluid phase in fracture and matrix are derived separately. These equations are discretized by the standard control-volume finite difference method (CVFD), and the flux is calculated through the two-point flux approximation (TPFA) scheme. The discretized mass conservation equation of phase α in the fracture (f) and matrix (m) system can be written as,

$$T_f \lambda_{\alpha,f}^{n+1}\left(\Delta p_{\alpha,f}^{n+1} - \rho_{\alpha,f}^{n+1} \mathrm{g}\Delta \mathrm{Z}\right) + T_{mf}\lambda_{\alpha,mf}^{n+1}\left(p_{\alpha,m}^{n+1} - p_{\alpha,f}^{n+1}\right) + q_\alpha^{n+1} - \frac{V_f}{\Delta t}\left[\left(\frac{\phi S_\alpha}{B_\alpha}\right)^{n+1}_f - \left(\frac{\phi S_\alpha}{B_\alpha}\right)^n_f\right] = 0 \tag{1}$$

$$T_m \lambda_{\alpha,m}^{n+1}\left(\Delta p_{\alpha,m}^{n+1}\right) + T_{mf}\lambda_{\alpha,mf}^{n+1}\left(p_{\alpha,f}^{n+1} - p_{\alpha,m}^{n+1}\right) - \frac{V_m}{\Delta t}\left[\left(\frac{\phi S_\alpha}{B_\alpha}\right)^{n+1}_m - \left(\frac{\phi S_\alpha}{B_\alpha}\right)^n_m\right] = 0 \tag{2}$$

where T_i represents the intra-continuum transmissibility internally in continuum i ($i = f, m$); T_{mf} is the inter-continuum transmissibility between matrix and fracture; $\lambda_{\alpha,i}$ is the phase mobility in continuum i; $\lambda_{\alpha, mf}$ is the inter-continuum mobility of phase α between matrix and fracture; $p_{\alpha,i}$ and $\rho_{\alpha, i}$ denote the phase pressure and density in continuum i; g is the gravity acceleration; Z is the node depth; q_α is the well flow rate perforated at the node; V_i is the node volume of continuum i; Δt is time step size; ϕ is porosity; S_α is phase saturation; B_α is the formation volume factor of phase α. The subscript α denotes the fluid phase, i.e., oil (o) or water (w); the superscripts n and $n+1$ represent the previous and current time steps, respectively.

The well flow rate of each phase α can be further calculated through the inflow model as represented in Eq. (3),

$$q_\alpha = \mathrm{WI} \cdot \lambda_\alpha^w\left(p_{\alpha,f}^w - p_{wf}\right) \tag{3}$$

where WI is the well index representing the well productivity or injectivity; λ_α^w is the fluid phase mobility in the well node; $p_{\alpha,f}^w$ is the phase pressure in the well node; p_{wf} is the well bottom-hole pressure.

Since the number of primary variables (e.g., water pressure and water saturation at each node) solved in DC-CGNet is usually in relatively small magnitude (100 to 1000), Eqs. (1) and (2) can be extremely efficiently solved through the fully implicit method based on traditional reservoir simulators, with CPU time at the level of seconds. Further details can be referred to (Ertekin et al., 2001)

History-Matching Method

During history matching, the tunable model parameters in DC-CGNet include fracture-to-fracture transmissibilities, fracture-to-matrix transmissibilities, matrix-to-matrix transmissibilities, pore volumes for matrix and fracture nodes, relative permeability curves, and well indices, and the number of model parameters in the DC-CGNet is in a relatively small dimension (100 to 1000). Rather than relying on the concept of shape factor in DPM, the inter-continuum (or matrix–to-fracture) transmissibilities are treated as freely tunable model parameters, providing greater flexibility in representing inter-continuum fluid transfer. In practice, we typically assign broader parameter ranges for fracture-to-fracture and matrix-to-fracture transmissibilities than that for matrix-to-matrix transmissibility, In contrast, the upper range for fracture pore volume is set lower than matrix pore volume. This is consistent with that fractures bring fluid migration capacity and matrices primarily offer pore space for fluid storage.

The Ensemble Smoother with Multiple Data Assimilation (ES-MDA) approach (Emerick & Reynolds, 2013) is adopted for history matching, since adjoint implementation is not often available in many existing reservoir simulators. ES-MDA is an iterative data assimilation approach. At each iteration n in ES-MDA, the model parameters in each ensemble member j is updated by Eq. (4),

$$\mathbf{m}_j^{n+1} = \mathbf{m}_j^n + \mathbf{C}_{MD}^n\left(\mathbf{C}_{DD}^n + \alpha_n \mathbf{C}_D\right)^{-1}\left(\mathbf{d}_{uc,j}^n - \mathbf{d}_j^n\right) \tag{4}$$

where α_n is the inflation factor, with $\alpha_n = 1/N_a$; N_a is the number of assimilation steps, chosen as 5; $\mathbf{C}_D$ is the error covariance matrix; $\mathbf{d}_{uc,j}^n$ denotes the perturbed observed data; $\mathbf{d}_j^n$ is the predicted data vectors; $\mathbf{C}_{MD}^n$ is the cross-covariance matrix between the parameters and the predicted data; $\mathbf{C}_{DD}^n$ is the autocovariance matrix of the predicted data, which are calculated using Eq. (5),

$$\begin{aligned} \mathbf{C}_{MD}^n &= \frac{1}{N_e - 1}\sum_{j=1}^{N_e}\left(\mathbf{m}_j^n - \overline{\mathbf{m}}^n\right)\left(\mathbf{d}_j^n - \overline{\mathbf{d}}^n\right)^T \\ \mathbf{C}_{DD}^n &= \frac{1}{N_e - 1}\sum_{j=1}^{N_e}\left(\mathbf{d}_j^n - \overline{\mathbf{d}}^n\right)\left(\mathbf{d}_j^n - \overline{\mathbf{d}}^n\right)^T \end{aligned} \tag{5}$$

Notice that in our implementation, all the DC-CGNet model parameters are scaled based on the min-max ranges of each parameter type to avoid scale imbalance in covariance matrix calculation.

Results of Numerical Experiments

The numerical experiments aim to validate the accuracy and effectiveness of the proposed hybrid surrogate model using a synthetic fractured reservoir represented by a high-resolution Embedded Discrete Fracture Model (EDFM) as the reference model. As a result, the EDFM model generates the historical and forecast reference well data used in the history matching. The well data here include well injection and production rates. During history matching, the tunable model parameters in DC-CGNet are history matched against the historical well data, and subsequently employed as an efficient surrogate model for dynamic well-control optimization. The performance of DC-CGNet is assessed in two aspects: (1) match the historical well data; (2) further forecast the well performance consistent with observational well data.

Description of Synthetic Naturally Fractured Reservoir Model

As EDFM can explicitly define the fracture geometry, location, and properties of individual fractures, it is used as a reference model to generate synthetic well data from a densely fractured reservoir. Figure 2 illustrates the reservoir grid discretization along with the well placements used in the reference EDFM. The reservoir domain is discretized using corner-point grid with 41 × 41 × 5 grid cells. A total of 100 fractures (yellow planes) are populated in the reservoir, with varying length, thickness, and orientation. There are eight production wells (green cones) along the center and ends of edge boundaries, and one injection well (blue cones) in the center, which form a 9-spot well pattern in the reservoir. No-flow (closed) boundary conditions is imposed on all the four edge boundaries.

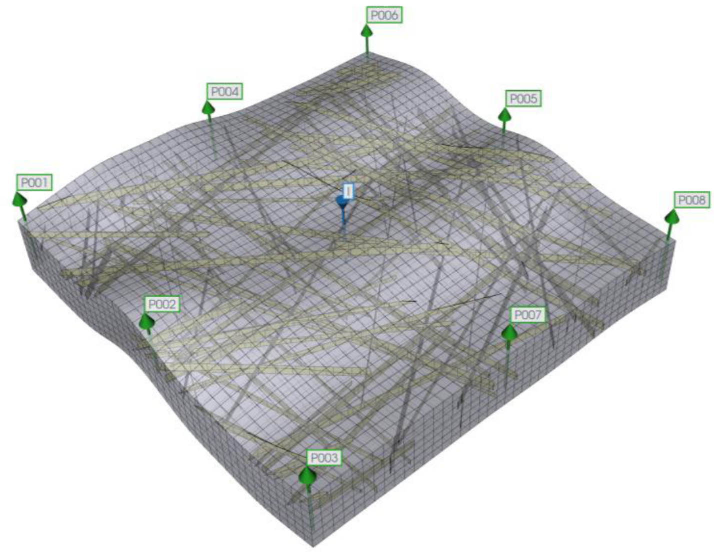


Figure 2—EDFM setup for a synthetic naturally fractured reservoir.

In the EDFM, the matrix domain has constant porosity of 0.1 and uniform permeability of 5 mD, while the fracture network has significantly higher permeability of 10000 mD to represent dominant flow pathways with fracture aperture of 0.3 ft. The water-flooding process is two-phase (oil-water) flow, and is simulated using the CMG-IMEX simulator (Computer Modelling Group Ltd., 2024). Table 1 summarizes other rock and fluid properties used in the simulation model. The reservoir is initialized with uniform reservoir pressure and oil saturation. As shown in Figure 3, two sets of relative permeability curves are separately defined for the matrix and fracture, in order to capture their difference in multiphase flow. Capillary pressure is ignored in the EDFM simulation.

Table 1—Reservoir model parameters used in the EDFM to generate synthetic well data.

Parameters	Values	Units
Reservoir size	1000 × 1000 × 25	ft^3
Grid number	41 × 41 × 5	
Rock compressibility, c_{rm},c_{rf}	3×10^{-6}	$psia^{-1}$
Matrix permeability, k_m	5	mD
Fracture permeability, k_f	10000	mD
Matrix porosity, ϕ_m	0.1	
Fracture aperture, e	0.3	ft
Water density, ρ_w	62.4	lb_m/ft^3
Water viscosity, μ_w	1.28	cP
Oil density, ρ_o	45.7	lb_m/ft^3
Oil viscosity, μ_{od}	1.21	cP
Initial oil saturation, S_O^{init}	0.95	Fraction
Initial reservoir pressure, p^{init}	4800	$psia$

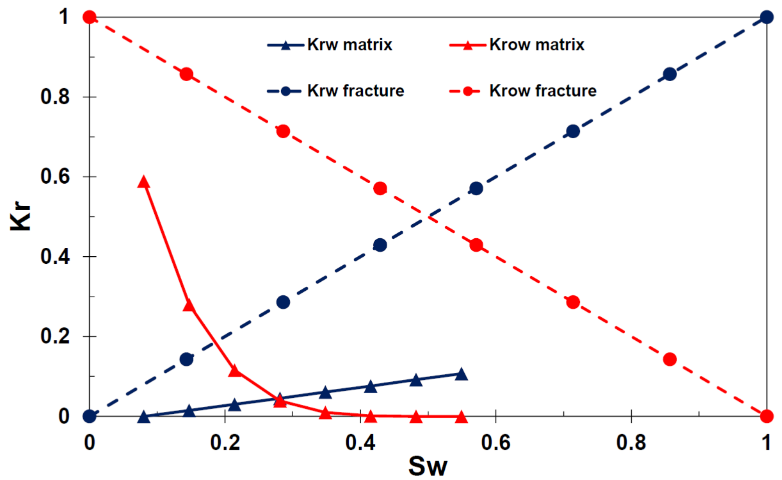


Figure 3—Fracture and matrix relative permeability curves used in the EDFM.

As shown in Figure 4, the synthetic reservoir based on the EDFM simulation is operated over a period of 3500 days. Besides, bottom-hole pressures (BHP) dynamically vary with time within specified limits: 100 to 250 psi for producers and 5000 to 7000 psi for the injector. These dynamic controls are applied at each time step, with each time step of size 180 days. After running the EDFM simulation, the synthetic well data include 17 time series data with a time span of 3500 days, including oil rates and water rates from eight production wells, and water injection rate from one injection well. Well data from the first 70% of the time span (0 to 2450 days) is used as historical data to perform history matching, and well data from the remaining 30% of the time span (2450 to 3500 days) is for forecast validation.

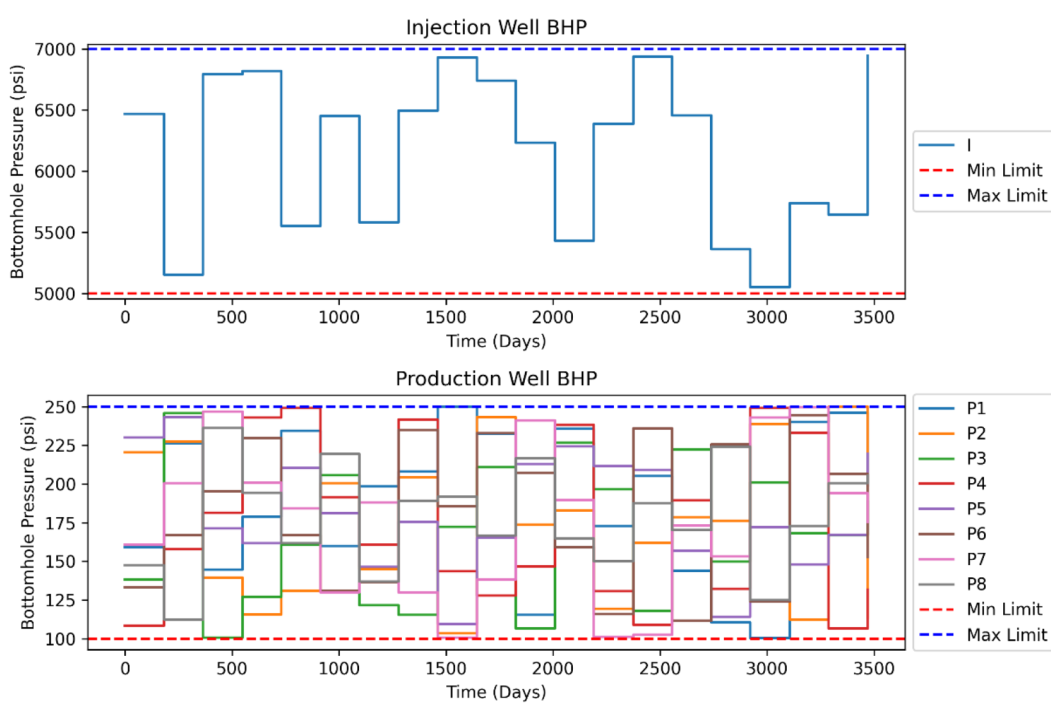


Figure 4—Injection well and production well control schedule of different wells in the synthetic reservoir. I*: injection well; P*: production well.

Sensitivity Analysis with DC-CGNet

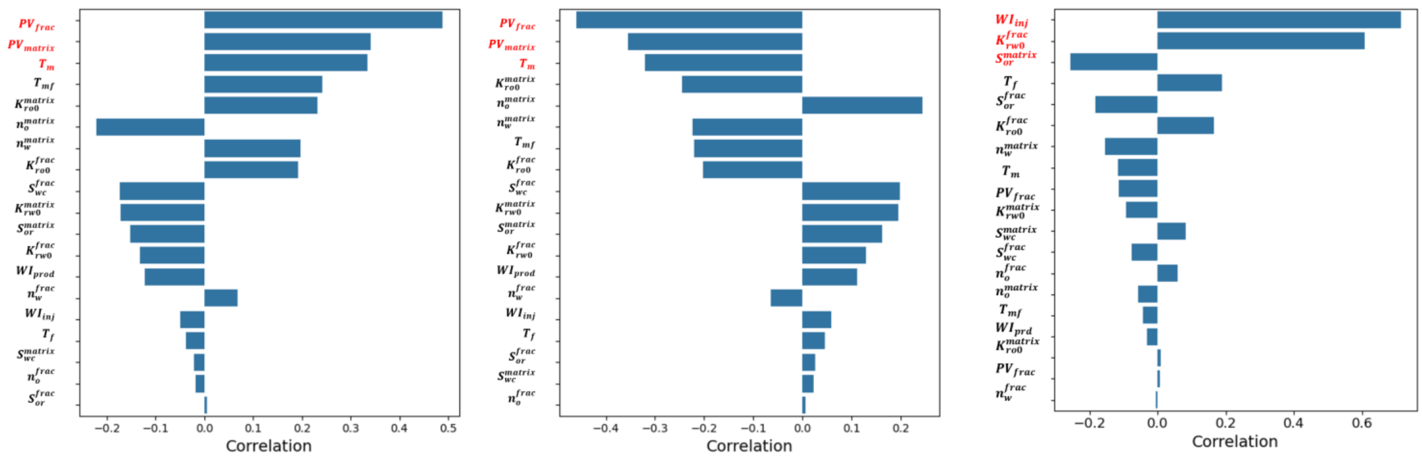
Sensitivity analysis was performed on key model parameters of DC-CGNet model to assess its effect on some quantity of interest (QoI) that describe the model responses such as well rates. The sensitivity analysis was also used to estimate the ranges of the model parameters by comparing the sampled DC-CGNet model responses to the synthetic well data. It is desired that the initial well responses from DC-CGNet to span the observational well data before it is used in ES-MDA calibration process. This also ensure better convergence and consistent result in ES-MDA calibration.

Table 2 lists the tunable model parameters of DC-CGNet along with their ranges and dimensions. There are in total 669 uncertain model parameters, and their prior distributions are assumed to be uniform, denoted as U(·,·). Following a sensitivity analysis of node resolution in DC-CGNet, 15 × 15 node resolution is selected for both matrix and fracture graph, which results in 225 nodes and 441 edges (connections) in both the fracture and matrix network. A prior ensemble consisting of 150 simulation samples ($n_{samples}$ = 150) is generated by sampling these parameter using Latin Hypercube Sampling (LHS) based on Table 2.

Table 2—Parameterization for DC-CGNet calibration.

Parameters	Uniform Distribution	Dimensions	Units
Fracture relative permeability endpoints, K_{ro0}^{frac}, K_{rw0}^{frac}	U(0.05, 1)	($n_{samples}$, 2)	–
Matrix relative permeability endpoints, K_{ro0}^{matrix}, K_{rw0}^{matrix}	U(0.05, 1)	($n_{samples}$, 2)	–
Fracture relative permeability exponent, n_o^{frac}, n_w^{frac}	U(1, 2)	($n_{samples}$, 2)	–
Matrix relative permeability exponent, n_o^{matrix}, n_w^{matrix}	U(1.01, 5.5)	($n_{samples}$, 2)	–
Fracture endpoint saturation, S_{or}^{frac}, S_{wc}^{frac}	U(0.01, 0.45)	($n_{samples}$, 2)	–
Matrix endpoint saturation, S_{or}^{matrix}, S_{wc}^{matrix}	U(0.01, 0.45)	($n_{samples}$, 2)	–
Fracture-to-fracture transmissibility, T_f	U(1000,10000)	($n_{samples}$, n_{edges}^{frac})	$md \cdot ft$
Matrix-to-matrix transmissibility, T_m	U(1, 1000)	($n_{samples}$, 1)	$md \cdot ft$
Fracture-to-matrix transmissibility, T_{mf}	U(1000, 1000000)	($n_{samples}$, 1)	$md \cdot ft$
Fracture node pore volume, PV_{frac}	U(100, 5000)	($n_{samples}$, n_{node}^{frac}	ft^3
Matrix node pore volume, PV_{matrix}	U(1000, 500000)	($n_{samples}$, 1)	ft^3
Producer well Index, WI_{prod}	U(10, 10000)	($n_{samples}$, n_{prod})	$md \cdot ft$
Injector well Index, WI_{inj}	U(10, 10000)	($n_{samples}$, n_{inj})	$md \cdot ft$

Figure 5 presents the Pearson correlation coefficients between the DC-CGNet model parameters and the selected QoIs, which include oil and water production rates, and injection water rate. As shown in Figures 5(a) and 5(b), the most dominant variables related to oil and water rates include the pore volume of fracture nodes (PV_{frac}), the pore volume of matrix nodes (PV_{matrix}), since the total fluid volume in the reservoir is controlled by PV_{frac} and PV_{matrix}. Further, the intra-continuum transmissibility for matrix (T_m) is also important to oil and water production rates. Even though T_m is significantly lower than T_f, the latter can contribute to the long-term production rates. As a result, the production rate of primary fluid phase (oil here) is positively correlated with these variables, while the water (secondary phase) production rate is negatively correlated with them. In contrast, for the injection rates, the injection well index (WI_{inj}) has the most significant influence, followed by the relative permeability parameters and then fracture transmissibility (T_f). This aligns with the well inflow model that governs the injectivity behavior, where increasing well index (injectivity), water mobility and fracture transmissibility leads to higher injection rate.



(a) Oil production rate (b) Water production rate (c) Injection water rate

Figure 5—Pearson correlation coefficients of the model parameters in DC-CGNet with respect to different QoIs. (a) oil rate; (b) water rate; (c) injection water rate.

History-Matching Result

The history matching (HM) was performed with the ES-MDA. The objective function during HM is the misfits between the predictions of DC-CGNet and the corresponding observational well data. To maintain the consistency and avoid introducing discrepancies between models due to well constraint switching, both production and injection wells are only controlled by bottom-hole pressure (BHP) controls, following the same schedule as the synthetic reservoir (Figure 4). The first 70% (2450 days) of the well data from the synthetic reservoir are used for history matching, which consist of 646 data points representing water injection rates, oil production rates, and water production rates at each well. As a result, the number of uncertain model parameters is 669, which is close to the number of observational data points (646), which favors the HM. The remaining 30% of the data was reserved for forecast verification. The number of ES-MDA iterations is 4, and the ensemble size is 150. Figure 6 shows the relative error of misfit against ES-MDA iterations. The prior DC-CGNet shows significant relative error (mean at around 200%) and high uncertainty bandwidth. After four iterations, the relative error drastically decreases with mean relative error at around 10%, and the uncertainty of the posterior model is at very low level. As a result, from the misfit point of view, ES-MDA is quite efficient to perform history matching for DC-CGNet.

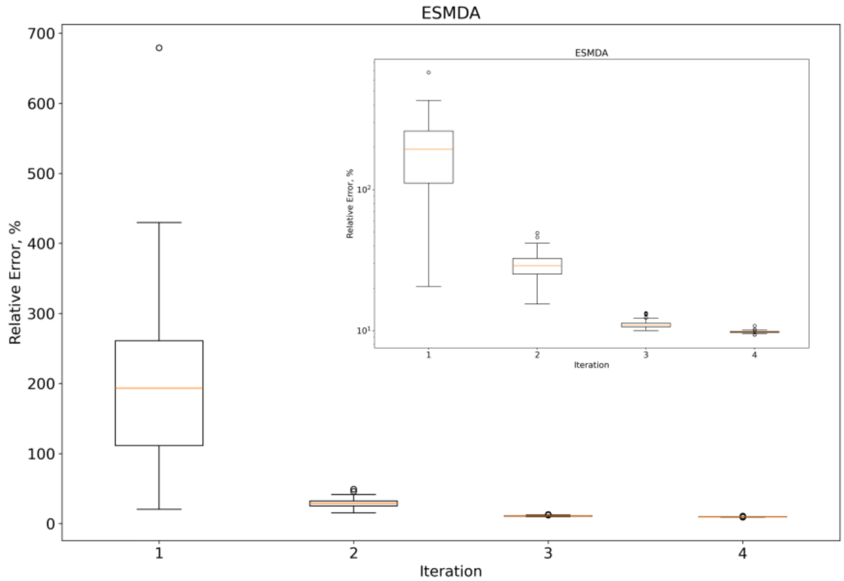


Figure 6—ES-MDA ensembles relative error to reference model output at each iteration (Inset: relative error in log scale).

Figures 7 to 9 compare the posterior ensemble responses from DC-CGNet with the observational well data. In general, the predictions from posterior DC-CGNet models show reasonable agreement with observational data well in all the nine wells, and the uncertainty in the posterior predictions are relatively low, especially in the water injection rate and oil production rates. In the injection well only single phase of water flows, and in the production well the oil phase is the dominant or primary fluid phase, such that the magnitude of water injection rate and oil production rates are relatively high and stable (10 to 100 STB/Day). This makes them relatively easier to history match. On the other hand, water phase is the secondary phase in production wells, and it only occurs after the water breakthroughs in these wells. As a result, the magnitude of water production rates cover a relatively larger range (0.01 to 10 STB/Day), which challenges the HM process. Further, the high uncertainty part in the water production rates mainly occurs in the early production period, which is shortly before or after the water breakthrough with no or extremely water production rates.

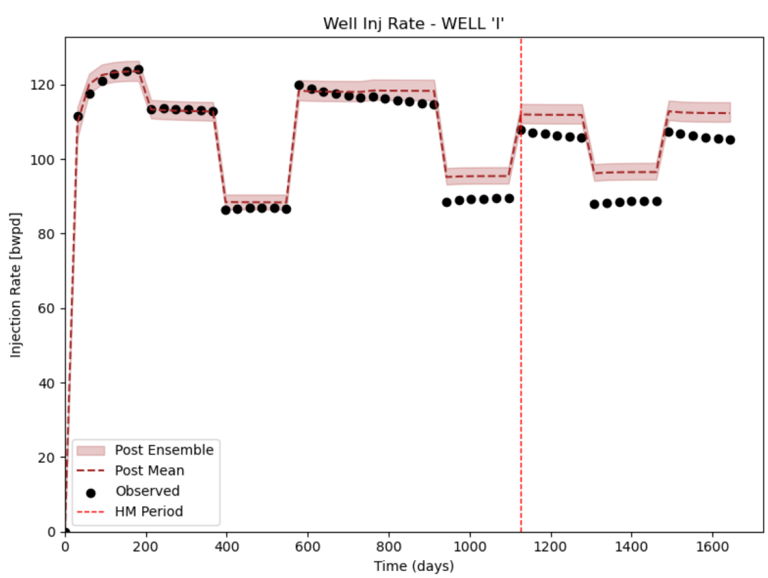


Figure 7—Comparison of water injection rates between the posterior ensemble (red) and the observational data (black circles). Red line separate the historical data (left) from the forecast data (right).

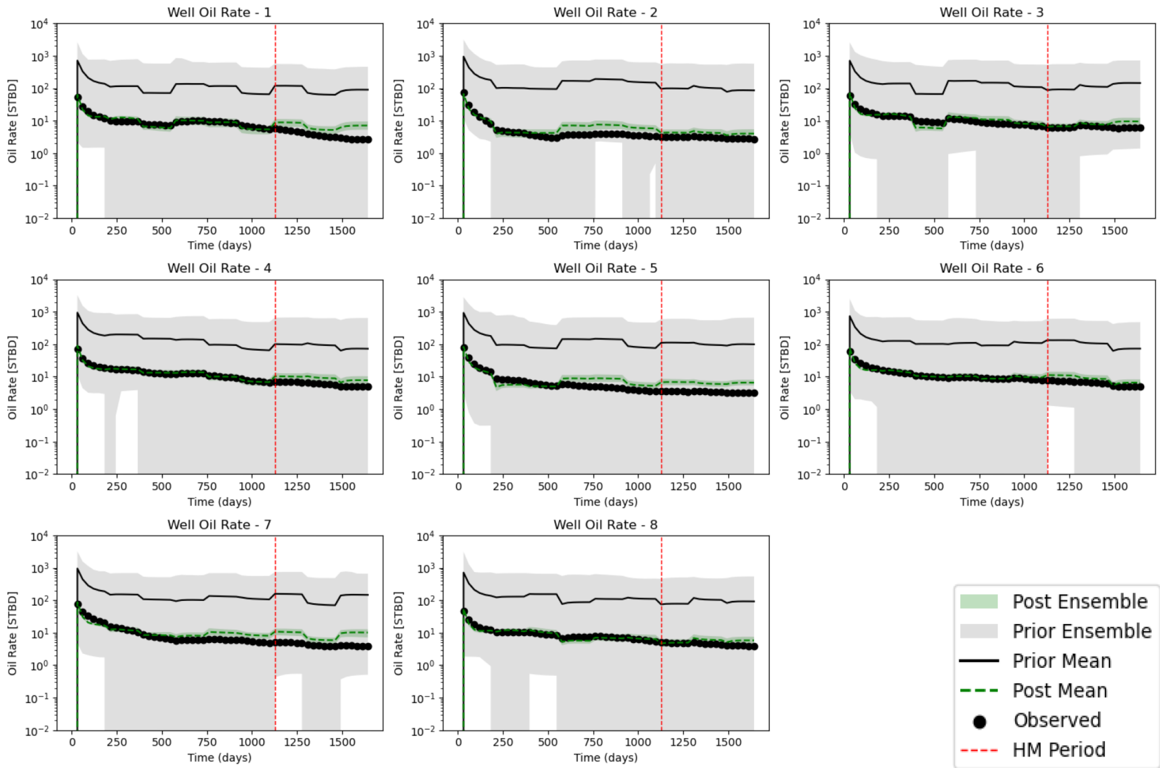


Figure 8—Comparison of oil production rates between the prior ensemble (grey), posterior ensemble (green), and the synthetic observational data (black). Red line separate the historical data (left) from the forecast data (right).

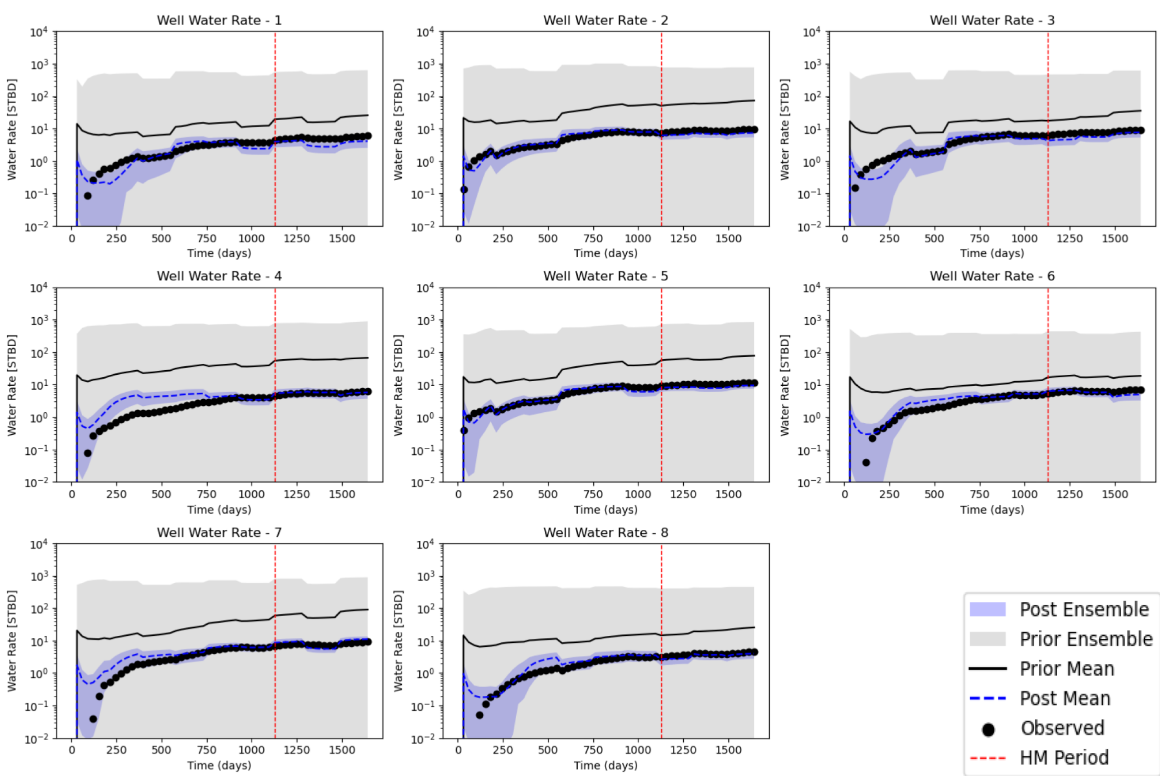


Figure 9—Comparison of water production rates between the prior ensemble (grey), posterior ensemble (blue), and the synthetic observational data (black). Red line separate the historical data (left) from the forecast data (right).

The satisfactory HM and forecast accuracy are favored by the model resolution and the parameterization strategies used. While increasing node resolution can help mitigate numerical smearing caused by coarse spatial discretization, it also introduces more parameters and higher computational cost. Based on our previous experience from CGNet (Aslam et al., 2023, 2024), carefully choosing which parameters to spatially vary or stay spatially uniform across nodes and edges can significantly influence the history-matching performance. As a result, the satisfactory HM performance in this study can be also benefited from the several adopted strategies, including: (1) assigning distinct relative permeability parameters to matrix and fracture graphs; (2) allowing fracture transmissibility to vary across fracture graph edges; (3) adding spatial variability in fracture pore volumes; (4) calibrating individual well indices. This customized parameterization strategy is also physically consistent, as densely fractured reservoir often lead to heterogeneous interactions among fracture, matrix, and well nodes. For instance, the density and orientation of inter-connected fractures can vary significantly across the domain, which can justify the use of locally varying fracture transmissibility, pore volume, and well index values.

Further, the DC-CGNet model maintains good predictive performance in the forecast stage, indicating that this hybrid surrogate model generalizes well to unseen data. This consistency increases the confidence to use DC-CGNet as a computationally efficient and physically informed surrogate to forecast fractured reservoir performance.

Conclusion

In this study, a Dual-Continuum Coarse Grid Network (DC-CGNet) was developed as a hybrid physics-based data-driven surrogate model. DC-CGNet excels at simulating, history matching, and optimizing densely fracture reservoirs, since it separately represents the fracture and matrix continua as inter-connected coarse graphs. DC-CGNet can flexibly couple with existing reservoir simulators. The number of model parameters in DC-CGNet is still small (~600 in this work), slightly two times more than the number of model parameters in CGNet, since reservoir properties such as relative permeability curves, transmissibility, and pore volumes need to be parameterized for both matrix and fracture, along with the well indices. DC-CGNet can further couple with Ensemble Smoother with Multiple Data Assimilation (ES-MDA) for history matching, which can quantify the uncertainty in the model parameters.

Based on a realistic case of water flooding process in a densely fractured reservoir, the numerical experiment shows that DC-CGNet can effectively accelerate fractured reservoir simulation with a speedup of 15 times. ES-MDA can effectively history match the reservoir production historical data with the prediction from DC-CGNet. Finally, the DC-CGNet model offers promising extensions to other physics processes in naturally fractured reservoirs, such as gas injection and enhanced geothermal systems, which will be in the next phase of the current research.

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